

PROBLEM STATEMENT #25 | Smart Cities, Transport & Logistics**Municipal Solid Waste Route Optimizer****1. Problem Context & Overview**

A city sanitation management portal that ingests ultrasonic fill-level data from smart trash bins, plans optimized collection routes for garbage trucks, and tracks missed-pickup citizen complaints.

Target Stakeholders / Actors: *Sanitation Supervisor, Truck Driver*

2. Sample Functional Requirement (FR) Guideline**Sample Requirement: FR-001 [Priority: High]**

Description: The system shall generate optimized daily collection routes including only bins that have exceeded an 80% fill threshold.

Sample Acceptance Criteria: Pass: Route excludes bins < 80% full unless manually tagged; Fail: Overflowing bin omitted from driver manifest.

3. Sample Non-Functional Requirement (NFR) Guideline**Sample Requirement: NFR-001 [Type: Performance & Security]**

Description: Route recalculation for a fleet of 50 trucks across 10,000 bins shall execute in under 45 seconds.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]**• Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

• UML Use-Case Diagram:

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

• Use-Case Flow Specification:

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.